

Plotting Data in R: Graphs

R has a number of packages for plotting graphs and data visualisation, such as graphics, lattice, and ggplot2. In this ninth article in the R series, we shall explore the various functions to plot data in R.



data.gov.in/catalog/all-india-consumer-price-index-ruralurban-0 for the different states in India. We can read the data from the downloaded file using the `read.csv` function, as shown below:

```
> cpi <- read.csv(file="CPI.csv", sep=",")

> head(cpi)
Sector Year Name Andhra.Pradesh Arunachal.Pradesh Assam Bihar
1 Rural 2011 January 104 NA 104 NA
2 Urban 2011 January 103 NA 103 NA
3 Rural+Urban 2011 January 103 NA 104 NA
4 Rural 2011 February 107 NA 105 NA
5 Urban 2011 February 106 NA 106 NA
6 Rural+Urban 2011 February 105 NA 105 NA
Chattisgarh Delhi Goa Gujarat Haryana Himachal.Pradesh
Jharkhand Karnataka
1 105 NA 103 104 104 104 105 104
2 104 NA 103 104 104 103 104 104
3 104 NA 103 104 104 104 103 105
104
4 107 NA 105 106 106 105 107 106
5 106 NA 105 107 107 105 107 108
6 105 NA 104 105 106 104 106 106
...
```

We will be using R version 4.1.2 installed on Parabola GNU/Linux-libre (x86-64) for the code snippets.

```
$ R --version
```

```
R version 4.1.2 (2021-11-01) -- "Bird Hippie"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)
```

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Plot

Consider the all-India consumer price index (CPI - rural/urban) data set up to November 2021 available at <https://data.gov.in/catalog/all-india-consumer-price-index-ruralurban-0>

Let us aggregate the CPI values per year for the state of Punjab, and plot a line chart using the `plot` function, as follows:

```
> punjab <- aggregate(x=cpi$Punjab, by=list(cpi$Year),
FUN=sum)

> head(punjab)
Group.1      x
1 2011 3881.76
2 2012 4183.30
3 2013 4368.40
4 2014 4455.50
5 2015 4584.30
6 2016 4715.80

> plot(punjab$Group.1, punjab$x, type="l", main="Punjab
Consumer Price Index upto November 2021", xlab="Year",
ylab="Consumer Price Index")
```